

Exercise 1

$$I \quad a^T \cdot x = \sum_{i=0}^{n-1} a_i x_i$$

$$\frac{\partial a^T x}{\partial x_j} = \frac{\partial \sum_{i=0}^{n-1} a_i x_i}{\partial x_j} = a_j$$

$$\rightarrow \frac{\partial a^T x}{\partial x} = a^T$$

II.

$$a^T A a = a^T \cdot \left(\sum_{i=0}^{n-1} \left(\sum_{j=0}^{n-1} A_{ij} \cdot a_j \right) \right) = \sum_{i=0}^{n-1} a_i \cdot \left(\sum_{j=0}^{n-1} A_{ij} a_j \right) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} a_i \cdot A_{ij} a_j$$

$$\frac{\partial a^T A a}{\partial a_k} = \frac{\partial \left(\sum_{i=0}^{n-1} \sum_{j=0}^{n-1} a_i A_{ij} a_j \right)}{\partial a_k} = \frac{\partial \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} A_{ij} \cdot (a_i a_j)}{\partial a_k} = \sum_{j=0}^{n-1} A_{kj} a_j + \sum_{i=0}^{n-1} A_{ik} a_i$$

$$\rightarrow \frac{\partial a^T A a}{\partial a} = \sum_{k=0}^{n-1} \frac{\partial a^T A a}{\partial a_k} = \sum_{k=0}^{n-1} \left(\sum_{j=0}^{n-1} A_{kj} a_j + \sum_{i=0}^{n-1} A_{ik} a_i \right) = \sum_{k=0}^{n-1} \sum_{j=0}^{n-1} A_{kj} a_j + \sum_{k=0}^{n-1} \sum_{i=0}^{n-1} A_{ik} a_i = a^T A^T + a^T A = a^T (A^T + A) = a^T (A + A^T)$$

III.

$$\frac{\partial (x - As)^T (x - As)}{\partial s_j} = \frac{\partial \left(\sum_{k=0}^{n-1} (x - As)_k (x - As)_k \right)}{\partial s_j} = \frac{\partial \sum_{k=0}^{n-1} x_k^2 - 2 x_k (As)_k + (As)_k^2}{\partial s_j} = \frac{\partial \sum_{k=0}^{n-1} (As)_k^2 - 2 x_k (As)_k}{\partial s_j}$$

$$= \frac{\partial \sum_{k=0}^{n-1} \left(\sum_{i=0}^{n-1} A_{ki} s_i \right)^2 - 2 x_k \left(\sum_{i=0}^{n-1} A_{ki} s_i \right)}{\partial s_j} = 2 \cdot \sum_{k=0}^{n-1} \sum_{i=0}^{n-1} A_{ki} s_i A_{kj} - 2 \cdot \sum_{k=0}^{n-1} x_k \cdot A_{kj} = \sum_{k=0}^{n-1} -2 \cdot \left(x_k A_{kj} - \sum_{i=0}^{n-1} A_{ki} s_i A_{kj} \right)$$

$$= -2 \sum_{k=0}^{n-1} \left(x_k - \sum_{i=0}^{n-1} A_{ki} s_i \right) A_{kj} = \left(-2 \cdot (x - As)^T A \right)_j$$

$$\frac{\partial (x - As)^T (x - As)}{\partial s} = \sum_{i=0}^{n-1} \frac{\partial (x - As)^T (x - As)}{\partial s_i} = \sum_{i=0}^{n-1} \left(-2 (x - As)^T A \right)_i = -2 (x - As)^T A$$

IV.

$$(x - As)^T A = (x^T - s^T A^T) \cdot A = x^T A - s^T A^T A$$

$$\frac{\partial x^T A - s^T A^T A}{\partial s_i} = - \frac{\partial s^T A^T A}{\partial s_i} = - \frac{\partial s^T B}{\partial s_i} = - \frac{\partial \left(\sum_{k=0}^{n-1} \sum_{j=0}^{n-1} s_i B_{jk} \right)}{\partial s_i} = \left(\sum_{k=0}^{n-1} B_{ik} \right)^T$$

$$\frac{\partial (-2 (x - As)^T A)}{\partial s} = 2 \sum_{i=0}^{n-1} \left(\sum_{k=0}^{n-1} B_{ik} \right)^T = 2 \sum_{i=0}^{n-1} \left(\sum_{k=0}^{n-1} (A^T A)_{ik} \right) = 2 A^T A$$